



29-INTEROP-AND-MIGRATION.md — Nobody abandons a 200,000-image Lightroom catalog



The realistic adoption path is not "switch to AURA". It is "use AURA for the brutal part, finish in Lightroom". Design for coexistence and you get adopted. Demand migration and you get evaluated once and abandoned.

1. The strategic position

AURA's competition is not Lightroom. It is the eight hours a photographer spends culling. Fit into their existing workflow at exactly two points and you are indispensable without asking anyone to change anything:

1. **Before Lightroom** — AURA culls 4,000 to 600 and hands over flags and ratings.
2. **Instead of the first edit pass** — AURA produces a develop starting point that Lightroom or Capture One can open and refine.

Everything else is optional. A photographer who never opens AURA's own editor is still a happy paying customer, and pretending otherwise leads to building an editor nobody asked for.

2. Interop surfaces

Surface	Direction	Fidelity	Priority
XMP sidecars	Both	Flags, ratings, colour labels, crop, basic tone — lossless for what it covers	P0, ship first
Lightroom catalog read	Import only	Read <code>.lrcat</code> (SQLite) for past cull decisions and edit history — pure gold for style learning	P0
Capture One session/catalog	Import only	Read ratings and adjustments from <code>.cosessiondb</code>	P1
DNG export	Export	Embedded adjustments, universally readable	P1
Folder + JPEG export	Export	The universal fallback that always works	P0
Photo Mechanic	Both	Via XMP; many pros cull here today, so match their expectations exactly	P1
Gallery hosts (Pic-Time, Pixieset, ShootProof)	Export	Structured folder export with the naming they expect	P2

3. XMP is a contract, not a convenience

XMP is how AURA talks to the rest of the industry, so it gets frozen-contract treatment.

```

<!-- schemas/sidecar.v1.xmp -- FROZEN MAPPING -->
<!-- Standard fields, read by everyone -->
xmp:Rating          -> 0-5, from AURA's keeper tiers
xmp:Label           -> colour label
crs:Exposure2012    -> tone.exposure_ev
crs:Contrast2012    -> tone.contrast
crs:Highlights2012  -> tone.highlights
crs:Shadows2012     -> tone.shadows
crs:Temperature     -> colour.temp_k
crs:Tint            -> colour.tint
crs:CropTop/Left/Bottom/Right -> crop rect, normalised

```

```
<!-- AURA namespace, ignored safely by other tools -->
aura:recipeVersion      -> recipe schema version
aura:profileVersion     -> style profile version (determinis
m!)
aura:confidence         -> 0.0-1.0
aura:decisionReason     -> human-readable, e.g. "eyes closed
3 of 4"
aura:runId              -> links back to the decision ledger
```

Rules:

- **Never destroy foreign fields.** Read, preserve, write back. A photographer who loses their Capture One adjustments because AURA rewrote the sidecar will tell every photographer they know.
- **Write atomically:** temp file, fsync, rename. A half-written sidecar next to an irreplaceable RAW is unacceptable.
- **Never write inside the original RAW.** Sidecars only, always.
- **Round-trip test is mandatory:** read a Lightroom sidecar, write it back untouched, assert byte-equivalent semantics.

4. Reading the Lightroom catalog

A `.lrcat` file is a SQLite database, and it contains the single most valuable dataset for style learning that exists: every keep and cut decision this photographer has ever made.

- **Open read-only, on a copy.** Never touch the original; a corrupted catalog ends the relationship permanently.
- **Extract:** pick flags and rejects, star ratings, colour labels, crop history, develop settings, collection membership, and export history.
- **The pairing:** match each RAW to the delivered JPEG through export history, and you have exactly the paired training data `28-STYLE-LEARNING-SPEC.md` needs, for free, from work already done.

- **Handle version drift:** the schema changes between Lightroom versions. Detect the version, support the common recent ones, and fail gracefully with a clear message rather than guessing at an unknown schema.
- **Ask before reading.** This is their entire professional history; the consent screen must be explicit and specific.

5. The migration ladder

Never ask for a leap. Offer four rungs, and let the photographer stop at any of them:

1. **Try** — one folder, sample wedding, nothing touched.
2. **Cull only** — AURA culls, exports XMP, Lightroom does everything else. This is where most users stay for months, and that is a success, not a failure.
3. **Cull plus develop starting point** — XMP includes tone and colour; they refine in their existing tool.
4. **Full AURA** — end to end, export direct to gallery.

Measure how many users sit on each rung. If nobody moves past rung 2, the editor is not good enough yet, and that is data, not defeat.

6. Failure modes

Failure	Consequence	Guard
Sidecar overwritten, foreign edits lost	Catastrophic trust loss	Preserve-unknown-fields test on real Lightroom and Capture One sidecars
Lightroom catalog written to	Corrupted professional archive	Open read-only on a copy; assert the original's hash is unchanged after import
Crop coordinate convention mismatch	Every crop is subtly wrong	Round-trip test with known crops across all three tools
Tone values interpreted differently	Exports look nothing like the preview	Visual parity corpus; document honestly where AURA and Lightroom differ instead of pretending they match

Failure	Consequence	Guard
Sidecar written mid-run, then the run resumes	Inconsistent state on disk	Sidecars written only at completion, atomically, idempotently
Unicode and long paths	Silent failure on Windows or on non-Latin filenames	Fixture set with Devanagari, CJK, emoji and 300-character paths

7. Tests

- ☐ **Preserve-unknown test:** sidecars from Lightroom, Capture One, Photo Mechanic and Bridge; write back; assert every foreign field survives byte-identically.
- ☐ **Read-only proof:** hash a `.lrcat` before and after import; assert unchanged.
- ☐ **Round-trip crop test:** known crops through AURA and back into each tool, verified visually and numerically.
- ☐ **Catalog version matrix:** at least four Lightroom Classic schema versions parse or fail with a clear message.
- ☐ **Atomicity test:** kill during sidecar write; assert the original sidecar is intact and no partial file remains.
- ☐ **Scale test:** import a 200,000-image catalog; measure duration and memory; must stay within the T-Typ ceiling.
- ☐ **Pairing accuracy:** on a known catalog, RAW-to-JPEG pairing accuracy above 95 %, with unmatched files reported rather than guessed.
- ☐ **Unicode fixture set** passes on all three platforms.

8. Acceptance criteria

- ☐ The XMP mapping in section 3 is frozen and under contract freeze.
- ☐ No code path can write to an original RAW or an original catalog; proven by architecture test.
- ☐ Foreign sidecar fields survive a full AURA round trip.

- ☐ Lightroom catalog import produces usable paired training data for style learning.
- ☐ All four migration rungs are reachable, and rung 2 is fully self-sufficient.
- ☐ Honest documentation exists comparing AURA and Lightroom rendering, including where they differ and why.
- ☐ A photographer with a 200,000-image catalog completed import successfully, timed and observed.

9. Claude Code prompt

Act as `MBE` (Backend) with `SEC` and `TLC` reviewing. Implement `29-INTEROP-AND-MIGRATION.md`: the frozen XMP mapping with full preservation of unknown fields, atomic sidecar writing, read-only Lightroom catalog import operating on a copy, the RAW-to-JPEG pairing extractor for style learning, and the Capture One reader.

Constraints: no write to any original RAW or original catalog anywhere in the codebase; sidecars written only at run completion, atomically and idempotently; unknown XMP fields must round-trip byte-identically; an unrecognised catalog schema fails with a clear coded error, never a guess.

Then run the preserve-unknown test against real sidecars from all four tools and paste the diffs, hash a `.lrcat` before and after import to prove read-only, and report pairing accuracy with the count of unmatched files.